compute the necessary length of minimum width copper trace (here are jlc's minimums) needed to equal 350 Ohm's of resistance.

Minimum Trace Width/Spacing (1 oz)	① 12 μm (0.33 oz) copper: 3/3 mil (absolute limit 2/2 mil – avoid if possible) ② 18 μm (0.5 oz) copper: 3.5/3.5 mil ③ 35 μm (1 oz) copper: 4/4 mil These are regular capabilities. Contact customer support for custom capability requirements.

For 12 micro meter copper , minimum width is 3 mil (absolute limit is 2).

Resistivity of copper at room temp is ρ = 1.68 ×10^-8 Ω m

We know that R = ρL/A

Since thickness is 12 micro meters and lets take width as 3 mil, then A = 7.62×10^{-5} m x 1.2×10^{-5} m = 9.144×10^{-10} m^2

If we rearrange the equation we get L = RA/ρ

So the length we need to get 350 ohm's of resistance is L = (350Ω)(9.144×10^{-10} m^2)/(1.68 ×10^-8 Ω m) = 19.05 meters

determine the size of rectangle needed to fit those traces according to JLC's published minimum trace width and trace clearance (width between traces).

We know that width is 3 mil and spacing is also 3 mil so to find total area of the rectangle to hold 19.05 meters of trace would just be 6 mil x 19.05 = 0.1524 mm x 19,050 mm = 2,902.32 mm^2 or 29.0232 cm^2

16 mm width restriction - so if area needs to be 2902 mm squared then length = 2902/16 mm =

So therefore dimensions of the rectangle is going to be 181 mm x 16 mm